

Abhishek Kumar Singh

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Professional Summary

Motivated B.Tech Computer Science student at BIT Mesra (**CGPA 8.91**) with hands-on AI/ML and data analytics experience. Built production-grade Power BI dashboards integrating Python ML models and NLP on real enterprise data. Developed and deployed an empathetic LLM counseling chatbot using LoRA/PEFT fine-tuning, FastAPI, and Docker on Hugging Face Spaces. Skilled across the full ML lifecycle — data wrangling, model training, evaluation, and cloud deployment. **GATE CS & DS 2026** qualifier.

Education

B.Tech — Computer Science and Engineering · CGPA: 8.91 2023 – 2027
Birla Institute of Technology, Mesra

Higher Secondary (10+2) — 94.6% 2020 – 2022
Holy Mary International School · Darbhanga, Bihar

Experience

Data Analytics Intern May 2026 – July 2026
Oil and Natural Gas Corporation Limited (ONGC) · Mumbai, Maharashtra

- Engineered a **5-page interactive Power BI dashboard** (Executive, SLA Analysis, Department Analytics, Sentiment Intelligence, AI Insights) processing **~2,000 real IT helpdesk tickets** for ONGC leadership.
- Integrated Python ML models — **Random Forest classifier** for priority prediction and **Gradient Boosting regressor** for resolution-time forecasting — directly into the Power BI pipeline via Python visuals.
- Performed end-to-end data wrangling in **Power Query (M language)**: null imputation, casing normalization, date parsing, SLA breach flag derivation; reduced data inconsistencies by **~35%**.
- Authored **25+ DAX measures** including SLA compliance rate, MTTR, ticket velocity, and agent performance index enabling real-time executive drill-through analysis.
- Applied **NLP-based sentiment scoring** (VADER / TextBlob) on ticket descriptions, surfacing department-level frustration trends used in monthly operations reviews.
- Delivered ONGC-branded reports reviewed and evaluated by senior mentors as the internship capstone deliverable.

Projects

EmpathAI — Empathetic LLM Counseling Chatbot 2026
Python · Hugging Face · LangChain · FastAPI · Docker · Streamlit · MLflow

- Fine-tuned **Mistral-7B using LoRA/PEFT** on curated mental-health conversation datasets, improving domain-specific empathy scores by **38%** over the base model in human evaluation.
- Designed a **multi-turn conversation pipeline with LangChain**, maintaining session context and implementing crisis-detection guardrails with safe fallback responses.
- Built a **REST API backend with FastAPI**, containerized with **Docker**, and deployed on **Hugging Face Spaces + Render**; achieved sub-2s inference latency on free-tier GPU.
- Validated with a 50-person pilot: **82% positive response quality** rating; reduced hallucination rate by **40%** vs. baseline through system-prompt engineering and **RAG (Retrieval-Augmented Generation)**.
- Implemented **MLflow experiment tracking** to log hyperparameters, loss curves, and ROUGE/BERTScore metrics across fine-tuning runs.

AI Code Review Bot — Automated PR Analysis Pipeline 2026
Python · FastAPI · Pydantic · LangChain · RAG · ChromaDB · AST · Ruff/Bandit · Pytest · GitHub

- Architected a modular **automated code-review pipeline** that fuses Git diff parsing, **AST-based structural analysis**, and deterministic static analysis (**Ruff, Bandit**) with LLM-based semantic reasoning to surface security, bug, performance, and code-quality findings on real code changes.
- Designed an **evidence-grounded RAG pipeline with ChromaDB**, indexing curated security, API, testing, and coding-standard knowledge so every AI-generated recommendation is retrieved and grounded rather than freely generated.
- Built a **FastAPI backend** with strict **Pydantic schemas** for request/response validation, an LLM-provider abstraction layer, structured JSON output parsing, and **confidence-based filtering** to keep findings reliable and actionable.
- Implemented a **review orchestrator** that validates, deduplicates, and severity-scores findings across the static-analysis, AST, and LLM layers into one unified, structured output.

- Wrote a comprehensive **Pytest suite** covering diff parsing, AST analysis, static analysis, RAG ingestion/retrieval, LLM response parsing, review validation, API error handling, and confidence-filtering edge cases.
- Scoped the system , extending the current pipeline toward GitHub App webhook integration, PostgreSQL persistence, a React review dashboard, and Docker-based CI/CD deployment.

Technical Skills

Languages: Python, Java, C, C++, SQL, JavaScript, HTML/CSS

ML / AI: Scikit-learn, Random Forest, Gradient Boosting, XGBoost, TensorFlow, PyTorch (basics), Transformers (Hugging Face), LoRA/PEFT fine-tuning, LangChain, RAG, Prompt Engineering

Code Analysis & Security: AST (Python `ast` module), Ruff, Bandit, Static Analysis, Git Diff Parsing

Data & BI: Power BI (DAX, Power Query M), Pandas, NumPy, Matplotlib, Jupyter Notebook

MLOps & Deployment: FastAPI, Pydantic, Docker, MLflow, Hugging Face Spaces, Render, Git, GitHub Actions (CI/CD basics), REST APIs, Pytest

Databases: MySQL, MongoDB, ChromaDB (Vector Store)

Tools & IDEs: VS Code, Anaconda, Google Colab

Achievements

- **GATE CS & DS 2026** — Qualified national-level exam, demonstrating strong proficiency in CS fundamentals, Data Structures, and Algorithms.
- Shortlisted for ONGC Data Analytics Internship (2025) from a competitive pool of CSE students across multiple universities.
- Published a literature survey on AI-driven mental health chatbots analyzing 20+ research papers on culturally sensitive LLM-based CBT approaches.
- **Sports Excellence** — Secured Runner-up in both Badminton and Cricket at inter-branch college tournaments, demonstrating teamwork and consistency.